Arithmetic Symmetry: From Prime Ideals to Modular Forms

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The equation

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

already hints at a problem with applying ordinary prime factorization outside the integers. In a number field the right objects for factorization are ideals; they allow one to examine arithmetic one prime at a time. Much further along that road, elliptic curves provide local numerical data that can be read both from Galois representations and from modular forms. This gives a concrete glimpse of why two such different theories meet.[1]

§1.1 Why elements stop having unique prime coordinates

In $R = \mathbb{Z}[\sqrt{-5}]$, the norm is

$$N(a + b\sqrt{-5}) = a^2 + 5b^2.$$

It is multiplicative and takes positive integer values on nonzero elements. In particular, there is no element of norm 2 or 3. This makes 2, 3, and $1 \pm \sqrt{-5}$ irreducible: any nontrivial factorization of one of these elements would force a factor of norm 2 or 3. The two factorizations of 6 above are genuinely distinct, since the only units in R are ± 1.

The ring R is the full ring of integers of $\mathbb{Q}(\sqrt{-5})$. Although elements do not factor uniquely, nonzero ideals do:

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(\mathfrak{a})}$$

with finitely many nonzero exponents. This is a theorem for rings of integers of number fields, which are Dedekind domains.

The ideal $\mathfrak{p} = (2, 1 + \sqrt{-5})$ is a tangible new arithmetic object. The quotient $R/\mathfrak{p}$ is $\mathbb{F}_2$, so $\mathfrak{p}$ is prime. Yet it is not principal: a generator would have absolute norm 2, and $a^2 + 5b^2 = 2$ has no integer solutions. The ideal class group records exactly this failure of every ideal to have a single generator. It is a finite abelian group for every number field.

§1.2 Local arithmetic records what happens at one prime

For a rational prime p, the p-adic absolute value measures divisibility:

$$|x|_p = p^{-v_p(x)} \qquad (x \neq 0).$$

Completing $\mathbb{Q}$ for the metric $|x - y|_p$ gives the field $\mathbb{Q}_p$. A sequence such as

$$1, \quad 1 + p, \quad 1 + p + p^2, \quad \dots$$

[1] The arithmetic context follows Ireland and Rosen, *A Classical Introduction to Modern Number Theory*, and Silverman, *The Arithmetic of Elliptic Curves*. The numerical modular-form data are recorded in the LMFDB level-32 isogeny class.

converges p-adically because the successive differences become divisible by increasingly high powers of p. Its limit is $1/(1-p)$ in $\mathbb{Q}_p$, even though the partial sums grow in the usual real metric.

For a number field K, completions K_v at its places include complex or real fields as well as finite extensions of $\mathbb{Q}_p$. Passing to K_v lets one study arithmetic near a chosen prime or infinite place. The local data do not automatically solve every global problem, but they are the natural inputs for Euler products and adelic descriptions.

Example 1.2.1 (Building a 7-adic square root, one digit at a time)
Although $\sqrt{2}$ is irrational over $\mathbb{Q}$, the equation $f(x) = x^2 - 2 = 0$ has a solution in $\mathbb{Q}_7$. The first digit is $x_1 = 3$, since $f(3) = 7 \equiv 0 \pmod 7$. To keep the first digit fixed while determining the next one, write $x_2 = 3 + 7t$. Then

$$f(3+7t) = 7 + 42t + 49t^2 \equiv 7(1+6t) \pmod{49}.$$

The correction must satisfy $1 + 6t \equiv 0 \pmod 7$, so $t \equiv 1 \pmod 7$. Hence

$$x_2 = 10, \qquad x_2^2 - 2 = 98 = 2 \cdot 49.$$

For a third digit let $x_3 = 10 + 49t$. Dividing $f(x_3)$ by 49 and reducing modulo 7 now gives

$$2 + 20t \equiv 2 + 6t \equiv 0 \pmod 7, \quad t \equiv 2 \pmod 7.$$

Thus

$$\boxed{3 \pmod 7, \quad 10 \pmod{49}, \quad 108 \pmod{343}}, \qquad 108^2 - 2 = 34 \cdot 343.$$

The general step is equally concrete. Given $f(x_k) \equiv 0 \pmod{p^k}$, put $x_{k+1} = x_k + p^k t$ and solve

$$\frac{f(x_k)}{p^k} + t f'(x_k) \equiv 0 \pmod p.$$

If $f'(x_k)$ is a unit modulo p, exactly one correction $t \bmod p$ works. The successive residue classes define a Cauchy sequence in $\mathbb{Z}_p$ and converge to a root: this is the simple-root case of Hensel's lemma.
There is no contradiction with irrationality over $\mathbb{Q}$: solvability in one completion is local information, not a rational solution of the original equation.

§1.3 An elliptic curve gives a computable local fingerprint

Consider the elliptic curve over $\mathbb{Q}$

$$E: \quad y^2 = x^3 - x.$$

Its discriminant is 64, so the reduction is smooth at every odd prime. For such a prime p, define

$$a_p = p + 1 - \#E(\mathbb{F}_p),$$

where the point at infinity is included in the count.

At $p = 5$, the values of $x^3 - x$ for $x = 0, 1, 2, 3, 4$ are

$$0, \quad 0, \quad 1, \quad 4, \quad 0 \pmod 5.$$

The nonzero squares $1, 4$ each admit two square roots in $\mathbb{F}_5$, while 0 admits one. Thus there are $1 + 1 + 2 + 2 + 1 = 7$ affine points and one point at infinity:

$$\#E(\mathbb{F}_5) = 8, \qquad \boxed{a_5 = 5 + 1 - 8 = -2.}$$

Hasse's theorem ensures $|a_p| \leq 2\sqrt{p}$ for every good prime. The point count is therefore a small integer deviation from the rough expectation $p + 1$.

§1.4 Finite Fields and Characteristic Polynomials

For a good prime p, the finite group $E(\mathbb{F}_p)$ is controlled by an arithmetic symmetry: the Frobenius map

$$(x, y) \longmapsto (x^p, y^p)$$

on the points over an algebraic closure of $\mathbb{F}_p$. Its fixed points are exactly the points defined over $\mathbb{F}_p$. This is why counting solutions of a cubic equation can reveal the trace of a linear operator.

To make the operator literal, choose a second prime $\ell \neq p$. The ℓ-adic Tate module $T_\ell(E)$, built by taking compatible ℓ^n-torsion points, is free of rank two over $\mathbb{Z}_\ell$. At good reduction the Frobenius action has characteristic polynomial

$$\boxed{P_p(T) = T^2 - a_p T + p}, \qquad a_p = p + 1 - \#E(\mathbb{F}_p).$$

This theorem includes two nontrivial assertions: the trace a_p is integral and independent of the auxiliary ℓ, and the determinant is p. Our count at $p = 5$ gives

$$P_5(T) = T^2 + 2T + 5.$$

Let its complex roots be α, β. Then

$$\alpha + \beta = -2, \qquad \alpha\beta = 5.$$

Over $\mathbb{F}_{p^r}$, the relevant Frobenius is the r-fold iterate, with eigenvalues α^r, β^r. Hence

$$\#E(\mathbb{F}_{p^r}) = p^r + 1 - (\alpha^r + \beta^r).$$

The point counts over *all* finite extensions are thereby encoded by one quadratic polynomial.

For example, with $p = 5$,

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-2)^2 - 10 = -6,$$

so

$$\boxed{\#E(\mathbb{F}_{25}) = 25 + 1 - (-6) = 32.}$$

The extension-field count required no enumeration of the 25^2 possible affine pairs. This also gives a useful sanity check on terminology: $\alpha^2 + \beta^2 = -6$ is the *Frobenius trace over* $\mathbb{F}_{25}$; it is distinct from the coefficient a_{25} of the modular form below.

Hasse's bound $|a_p| \leq 2\sqrt{p}$ says the Frobenius roots have complex absolute value $\sqrt{p}$. Thus even the approximate count $p + 1 + O(\sqrt{p})$ reflects a spectral statement.

§1.5 A modular form with the same Euler factors

The Hasse–Weil L-function packages the point-counting data into local factors. At good primes,

$$L_p(E,s) = \frac{1}{1 - a_p p^{-s} + p^{1-2s}}.$$

Writing $X = p^{-s}$, the denominator is $1 - a_p X + pX^2$. Its reciprocal gives a recurrence:

$$\frac{1}{1 - a_p X + pX^2} = 1 + a_p X + (a_p^2 - p)X^2 + \cdots.$$

For our curve at $p = 5$, this predicts the coefficients

$$a_5 = -2, \qquad a_{25} = a_5^2 - 5 = -1.$$

Compare the latter with the trace -6 over $\mathbb{F}_{25}$. The two quantities are related by $a_{25} = (\alpha^2 + \beta^2) + p$, but they answer different questions: one is a Dirichlet-series coefficient, the other a trace of Frobenius squared.

The curve $E : y^2 = x^3 - x$ has conductor 32. The modularity theorem associates it with a normalized weight-two newform for $\Gamma_0(32)$. In this particular case the newform has the remarkably concrete product expression

$$\boxed{f(z) = \eta(4z)^2\eta(8z)^2 = q\prod_{n=1}^{\infty}(1-q^{4n})^2(1-q^{8n})^2}, \qquad q = e^{2\pi i z},$$

where $\eta(z) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ is the Dedekind eta function. Expanding the first few factors yields

$$f(z) = q - 2q^5 - 3q^9 + 6q^{13} + 2q^{17} - q^{25} + \cdots.$$

The coefficients $a_5 = -2$ and $a_{25} = -1$ agree exactly with the predictions from the elliptic curve.

What makes this an arithmetic theorem rather than a numerical coincidence is the equality of L-functions:

$$\boxed{L(E,s) = L(f,s).}$$

Equivalently, the Hecke eigenvalue $a_p(f)$ of the modular form matches $p+1-\#E(\mathbb{F}_p)$ at each good prime. The existence of a modular form with this property for every elliptic curve over $\mathbb{Q}$ is the deep content of modularity; the eta product allows us to inspect one instance term by term.

§1.6 Complex Multiplication and Symmetry

The curve has a particularly useful extra symmetry over $\mathbb{Q}(i)$:

$$(x, y) \longmapsto (-x, iy).$$

It respects the curve equation because

$$(iy)^2 = -y^2 = (-x)^3 - (-x).$$

Together with the usual integer multiplication maps, this gives complex multiplication by $\mathbb{Z}[i]$. At primes $p \equiv 3 \pmod 4$, a shadow of this symmetry is already visible in elementary arithmetic.

Let χ be the quadratic character on $\mathbb{F}_p$, with $\chi(0) = 0$. For each $x \in \mathbb{F}_p$, the equation $y^2 = x^3 - x$ has $1 + \chi(x^3 - x)$ solutions in y. Including the point at infinity, we get

$$\#E(\mathbb{F}_p) = p + 1 + \sum_{x \in \mathbb{F}_p} \chi(x^3 - x), \qquad a_p = -\sum_{x \in \mathbb{F}_p} \chi(x^3 - x).$$

Now $x^3 - x$ is odd in x. If $p \equiv 3 \pmod 4$, then $\chi(-1) = -1$, so the terms paired at x and $-x$ cancel:

$$\chi((-x)^3 - (-x)) = \chi(-1)\chi(x^3 - x) = -\chi(x^3 - x).$$

Consequently

$$\boxed{p \equiv 3 \pmod 4 \quad \Longrightarrow \quad a_p = 0.}$$

This predicts $a_3 = a_7 = a_{11} = 0$ without counting individual points. The modular form displays the same pattern: its eta product has only exponents $n \equiv 1 \pmod 4$, so its Fourier coefficients at indices $3, 7, 11, \ldots$ vanish.

There is a natural progression of ideas here. The factorization of ideals in a number field retains arithmetic data that elements can lose. Finite-field point counts extract local invariants of an elliptic curve. The Frobenius polynomial organizes these invariants over extension fields; the Euler factors collect them prime by prime. Modularity then places the same coefficients inside a holomorphic form, while complex multiplication explains a conspicuous pattern of vanishing coefficients on both sides. The general Langlands philosophy extends such matching of Galois and automorphic data, but the curve above already exhibits an explicit and verifiable instance.